COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	07/09/2026	**DATE SUBMITTED:**	07/14/2026
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ACTIVITY NO. 1 [REVIEW OF C++ PROGRAMMING]

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a function prototype. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Note:

- The driver program must call each function.
- The definitions must be after the main function.

Sections	Answer
Header File Declaration Section	`#include <iostream>`

Global Declaration Section	`int sumOfTwoNum(int fNum, int sNum);`
	`void sumOfThreeNum(int fNum, int sNum, int tNum);`
	`void grtrVar (int fNum, int sNum);`
	`bool logicalOperation (bool a, bool b);`

Class Declaration and Method Definition Section	

Main Function

```cpp
int main(){
    // FIRST BULLET
    std::cout << "The sum of the two numbers is: " << sumOfTwoNum(6,7) << std::endl;;
    sumOfThreeNum(6,7,8);
    // SECOND BULLET
    grtrVar (50,49);
    // THIRD BULLET
    logicalOperation (true, false);
}
```

Method Definition

```cpp
int sumOfTwoNum(int fNum, int sNum){

    int sum = fNum + sNum;

    return sum;
}

void sumOfThreeNum(int fNum, int sNum, int tNum){
    std::cout << "The total sum is: " << fNum + sNum + tNum << std::endl;
}

void grtrVar (int fNum, int sNum){
    if (fNum > sNum){
        std::cout<< "The number " << fNum <<" is greater than " << sNum << std::endl;
    }
    else if ( fNum < sNum){
        std::cout<< "The number " << fNum <<" is not greater than " << sNum << std::endl;
    }
    else if ( fNum = sNum){
        std::cout<< "The number " << fNum <<" and " << sNum <<" is equall "<< std::endl;
    }

}

 bool logicalOperation (bool a, bool b){
    std::cout << "AND OPERATOR: ";

    if(a && b){
        std::cout << " TRUE "<< std::endl;
    }
    else{
        std::cout << " FALSE "<< std::endl;
    }

    std::cout << "OR OPERATOR: ";

    if(a || b){
        std::cout << " TRUE "<< std::endl;
    }
    else{
        std::cout << " FALSE "<< std::endl;
    }
    std::cout << "NOT A: ";

    if(!a){
        std::cout << " TRUE "<< std::endl;
    }
    else{
        std::cout << " FALSE "<< std::endl;
    }

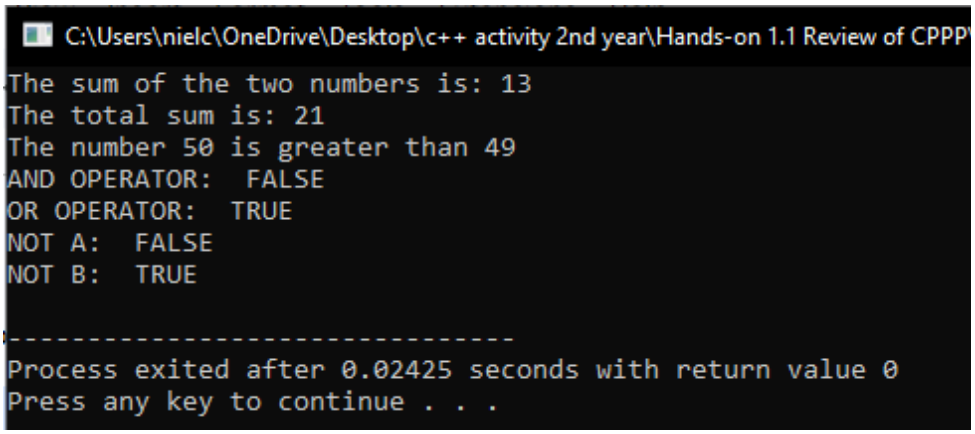
    std::cout << "NOT B: ";

    if(!b){
        std::cout << " TRUE "<< std::endl;
    }
    else{
        std::cout << " FALSE "<< std::endl;
    }


    return 0;
}
```

C++ Structure Code for Answer

OUTPUT:



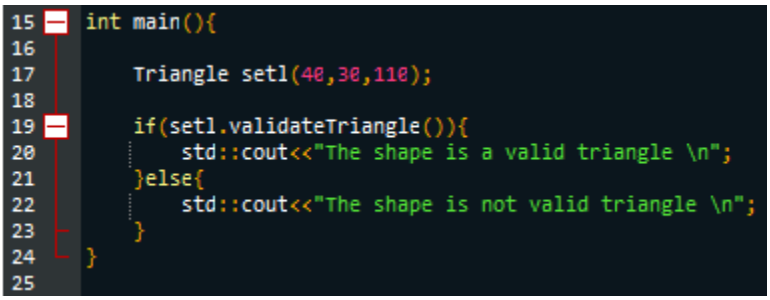
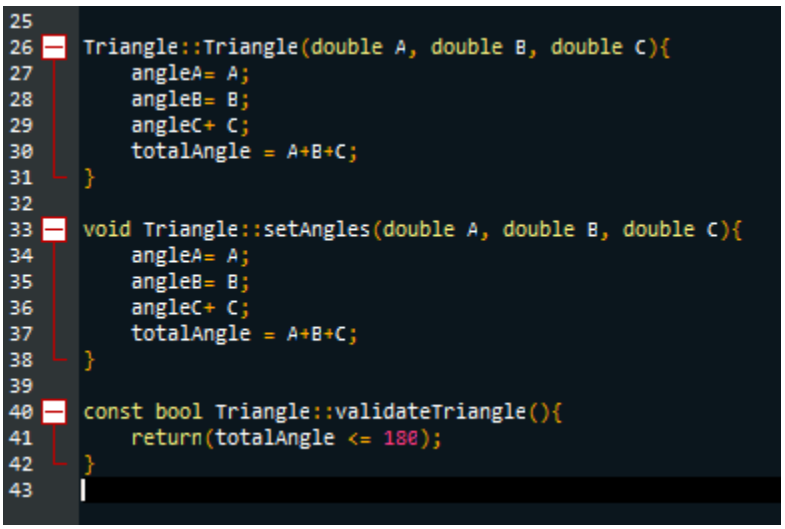
This program uses functions, It calculates and displays the sum of two numbers and three numbers. Next, it compares the two integers whether the first number is greater than, less than or equal to the second number. The last one is logical operation and, or, not on two boolean values and displays True or False results.

output observations and comments.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`// class name PASCAL CASE`
	`class Triangle{`
	`  private:`
	`  double totalAngle, angleA, angleB, angleC;`
	`  public:`
	`  //constructor`
	`  Triangle(double A, double B, double C);`
	`  void setAngles(double A, double B, double C);`
	`  const bool validateTriangle();`
	
	`};`

Main Function	
Method Definition	

```int main(){

    Triangle setl(40,30,110);

    if(setl.validateTriangle()){
        std::cout<<"The shape is a valid triangle \n";
    }else{
        std::cout<<"The shape is not valid triangle \n";
    }
}
```

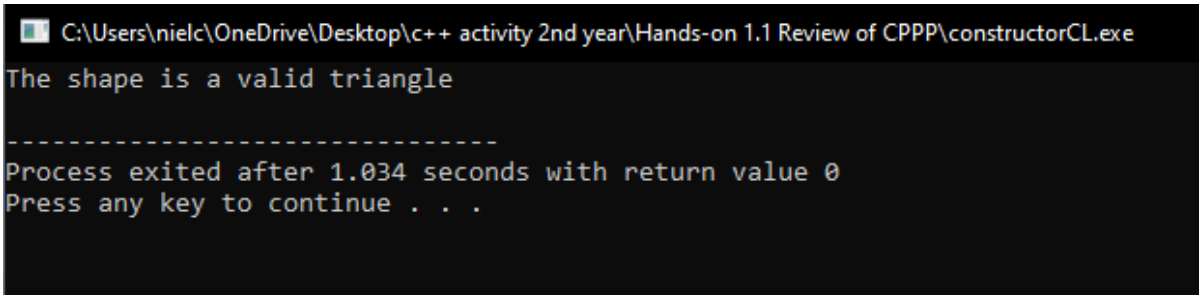
```Triangle::Triangle(double A, double B, double C){
    angleA= A;
    angleB= B;
    angleC+ C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA= A;
    angleB= B;
    angleC+ C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle(){
    return(totalAngle <= 180);
}
```

C++ Structure Code for Answer

OUTPUT:



This program uses a class,It creates a triangle with angles of **40°, 30°, and 110°**. It adds the three angles together and gets a total of **180°**. Since the total meets the condition set in the program, it prints the message **"The shape is a valid triangle."**

output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.

CODE:

```cpp
#include <iostream>

void swapNumbers(int a, int b) {
    int temp;

    temp = a;
    a = b;
    b = temp;

    std::cout << "\nAfter Swapping:\n";
    std::cout << "First Number = " << a << std::endl;
    std::cout << "Second Number = " << b << std::endl;
}

int main() {
    int num1, num2;

    std::cout << "Enter first number: ";
    std::cin >> num1;

    std::cout << "Enter second number: ";
    std::cin >> num2;

    swapNumbers(num1, num2);

    return 0;
}
// ANALYSIS: This program asks the user to enter two numbers and swaps them
//The swapped values are only shown inside the function
// because the original numbers in main() do not change.
```

OUTPUT:

```
Enter first number: 10
Enter second number: 25

After Swapping:
First Number = 25
Second Number = 10

--------------------------------
Process exited after 25.61 seconds with return value 0
Press any key to continue . . .
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
CODE:

```cpp
#include <iostream>

double KelvinToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9 / 5 + 32;
}

int main() {
    double kelvin;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> kelvin;

    std::cout << "Temperature in Fahrenheit = "
              << KelvinToFahrenheit(kelvin) << std::endl;

    return 0;
}
// ANALYSIS: This program asks the user to enter a temperature in Kelvin
// and converts it to Fahrenheit using a function. It then displays the
// converted temperature on the screen.
```

OUTPUT:

```
Enter temperature in Kelvin: 250
Temperature in Fahrenheit = -9.67

--------------------------------
Process exited after 11.76 seconds with return value 0
Press any key to continue . . .
```

3. Create a C++ program that has a function that will calculate the distance between two points.
CODE:

```cpp
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1, double x2, double y2) {
    return std::sqrt(std::pow(x2 - x1, 2) + std::pow(y2 - y1, 2));
}

int main() {
    double x1, y1, x2, y2;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    std::cout << "Distance = "
              << calculateDistance(x1, y1, x2, y2) << std::endl;

    return 0;
}
// ANALYSIS: This program asks the user to enter the coordinates of two points
// and uses the calculateDistance() function to find the distance between them
// It then displays the calculated distance on the screen.
```

OUTPUT:

```
Enter x1: 2
Enter y1: 4
Enter x2: 6
Enter y2: 8
Distance = 5.65685

--------------------------------
Process exited after 18.99 seconds with return value 0
Press any key to continue . . .
```

4. Modify the code given in ILO B and add the following functions:
    a. A function to compute for the area of a triangle
    b. A function to compute for the perimeter of a triangle
    c. A function that determines whether the triangle is acute-angled, obtuse-angled or ‘others.

CODE:

```cpp
#include <iostream>
#include <cmath>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C,
             double sA, double sB, double sC);

    void setAngles(double A, double B, double C);
    bool validateTriangle();

    double computePerimeter();
    double computeArea();
    void triangleType();
};

int main() {

    Triangle t1(40, 30, 110, 5, 6, 7);

    if (t1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";

        std::cout << "Perimeter: " << t1.computePerimeter() << std::endl;
        std::cout << "Area: " << t1.computeArea() << std::endl;

        std::cout << "Triangle Type: ";
        t1.triangleType();
    }
    else {
        std::cout << "The shape is not a valid triangle.\n";
    }

    return 0;
}

```

```cpp
Triangle::Triangle(double A, double B, double C,
                   double sA, double sB, double sC) {

    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;

    sideA = sA;
    sideB = sB;
    sideC = sC;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

// B. Compute the perimeter of a triangle
double Triangle::computePerimeter() {
    return sideA + sideB + sideC;
}

// A. Compute the Area of the Triangle
double Triangle::computeArea() {
    double s = computePerimeter() / 2;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

// C. Determine Triangle if Acute, Obtuse or Other
void Triangle::triangleType() {

    if (angleA < 90 && angleB < 90 && angleC < 90)
        std::cout << "Acute Triangle";
    else if (angleA > 90 || angleB > 90 || angleC > 90)
        std::cout << "Obtuse Triangle";
    else
        std::cout << "Others (Right Triangle)";
}
```

OUTPUT:

```
The shape is a valid triangle.
Perimeter: 18
Area: 14.6969
Triangle Type: Obtuse Triangle
--------------------------------
Process exited after 0.01337 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS: This program uses a Triangle class with functions to validate a triangle, calculate its perimeter, find its area using Heron's Formula, and classify it as an acute, obtuse, or right triangle based on its angles. It then displays all the computed results.

CONCLUSION: In this activity, we reviewed the previous lessons about the use of functions in c++. These exercises reviewed how to create functions for the task.This activity helped refresh my understanding of C++ and how functions can be used to create organized and efficient programs. It would be a warm up for the real tasks programming challenges that we will do in the future.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGpt, AI Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concepts, grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use AI for code assistance like how to use functions properly, the concept of cmath library and some grammar correction.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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